



Khaled Jallouli

Senior Software Engineer & Technical Lead at PreciTaste Inc.

Personal Information

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Experience

Since Apr. **Senior Software Engineer, PreciTaste Inc.**, Munich, Germany

- 2021
- Tech lead of three projects, managing a team of 15 (Design, Backend, Frontend, and QA).
 - Align priorities with Customer Success and define functional specs.
 - Plan monthly release scopes based on technical feasibility and team capacity.
 - Work on the project architecture whenever a new project is added.
 - Work closely with the Design team to define and validate UI/UX changes.
 - Create epics and Jira tickets to track feature development, bug fixes, and technical tasks.
 - Contribute to technical documentation in Confluence, providing specifications and feature definitions.
 - Architect RESTful APIs with Swagger.
 - Implement microservices within cloud environments using Django and FastAPI.
 - Define Kafka Streams data and topics utilizing Faust as a stream processing library.
 - Build aggregation pipelines in MongoDB to optimize performance.
 - Apply Python typing and Mypy for better code quality.
 - Deploy applications with Docker, Podman, and Kubernetes.
 - Follow trunk-based Git workflows and CI/CD practices.
 - Enforce PEP8 with Flake8 and Black for consistent style.
 - Write unit tests using Pytest as well as performance and load tests using JMeter.
 - Support in Integration Testing and User Acceptance Testing.
 - Ensure data security and risk assessment per ISO 27001.
 - Participate in the Software Audit.
 - Conduct code reviews, mentor junior engineers and working students, and engage in pair programming to ensure code quality and knowledge sharing.
 - Participate in technical interviews for new joiners.
 - Used technologies: Python · FastAPI · Django · Microservices · Kubernetes · Apache Kafka · MongoDB · Restful API · WebSocket · ISO 27001 · Agile Methodologies · pytest · Docker · Object-Oriented Programming (OOP) · Unified Modeling Language (UML) · Pair Programming · Swagger API · Postman API · Nginx · Celery · Slack · Jira · Git · NoSQL · JSON · Scrum · Unit Testing · Linux · JMeter

Augsburg, Germany

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- Since Oct. 2025 **Founding Software Engineer (Pre-Launch), Gym Tracking App**
- Defined the product vision for a gym tracking and AI-assisted training platform.
 - Designed the architecture across iOS and backend services.
 - Created UX designs in Figma.
 - Built a backend using Python (FastAPI) with RESTful APIs and MongoDB.
 - Implemented CI/CD pipelines with GitLab to automatically deploy backend services to Google Cloud.
 - Developed the iOS application in Swift, applying Domain-Driven Design (DDD) principles.
 - Integrated Fastlane and TestFlight for automated builds and internal testing.
 - Enrolled in the Apple Developer Program to support production iOS deployment.
 - Integrated LangChain-based AI services to generate personalized workout improvement suggestions.
 - Designed the platform with extensibility in mind to support future AI coaching and nutrition tracking features.
- Since Mar. 2020 **Private tutor, Superprof.de, Germany**
- Provide online/in-person tutoring to students of different nationalities, backgrounds and ages.
 - Help students understand concepts related to their studies or work on personal projects.
 - Teach lessons related to mathematics, programming, data science, and web development.
- Aug. 2020 – Feb. 2021 **Data Science Intern (Master Thesis), KAMAG Transporttechnik, Ulm, Germany**
- Optimization of yard movements and automatic generation of transport orders.
 - Find the most appropriate approach for the purpose of the project based on related works and research.
 - Analyze the Yard Management System data and constraints to define optimization strategies.
 - Model and solve the optimization strategies to reduce the driven distance in the yard.
 - Used technologies: Data Analysis, Constraint Programming, Optimization, IBM ILOG CPLEX, Data Mining Methodology, Python, RESTful API, Swagger, Flask, Vue.js, GitHub, and LaTeX.
- Feb. 2019 – Jul. 2019 **Software Engineer Intern, CP Georges Renault, Nantes, France**
- Setting up a server and developing a web interface to make the configuration of a tightening controller.
 - Design a web architecture that meets company constraints.
 - Develop the back-end and the front-end, then integrates them into the tightening controller.
 - Used technologies: RESTful API, Cutelyst (C++), Vue.js, JSON, NGINX, Responsive Web Design (Adobe XD), Scrum, and SVN.
- Jun. 2018 – Aug. 2018 **Web Development Intern, Leads Generation, Concarneau, France**
- Development of a website that references existing Business-to-Business events.
 - Define the main functionalities and implement a responsive website.
 - Used technologies: Drupal 7, HTML 5, CSS 3, PHP, Twig, JavaScript, JQuery, OVH, and Mailchimp.
- Jun. 2017 – Aug. 2017 **Web Development Intern, BR&T, Tunis, Tunisia**
- Development of a website presenting the start-up CCA Tunisia.
 - Install CMS plugins and implement a responsive website.
 - Used technologies: WordPress, HTML 5, CSS 3, PHP, and Mailjet.
- Jan. 2017 – May 2017 **Web Development Intern, Arzen Systems, Ankara, Turkey**
- Development of a learning management platform to manage courses, submit assignments and check grades.
 - Define the functionalities using UML diagrams and develop a web platform.
 - Used technologies: UML, PHP, MySQL, JavaScript, HTML 5, CSS 3, Bootstrap, and AJAX.

Academic Projects

- Mar. 2020 – Jun. 2020 **Computer Vision, Technische Hochschule, Ulm, Germany**
- Developed a mobile robot system to detect indoor traffic signs from video streams.
 - Evaluated transfer learning approaches for object detection using Google Coral.
 - Technologies: Deep Convolutional Neural Networks, Single Shot Detection (SSD), Transfer Learning, Google Coral TPU, Raspberry Pi 4, LaTeX.
- Oct. 2019 – Jun. 2020 **Machine Learning, Technische Hochschule, Ulm, Germany**
- Built a predictive model to forecast match outcomes based on teams' last ten games.
 - Applied supervised learning algorithms for classification (Decision Trees, Neural Networks).
 - Technologies: Python, TensorFlow, Keras, Scikit-learn, Pandas, Git, Scrum, CRISP-DM, LaTeX.
- Oct. 2018 – Dec. 2018 **Sensor Data Acquisition, National School of Engineering, Brest, France**
- Designed a human-machine interface to switch between light and distance sensors and visualize sensor data.
 - Technologies: Cortex-M3 Microcontroller, C, C++ (Qt), OpenGL, CAN Bus.

- Oct. 2018 – **Robot Control**, *National School of Engineering*, Brest, France
 Dec. 2018 Implemented remote robot control via a human-machine interface.
 ○ Technologies: STM32 Microcontroller, Ultrasonic and IR Sensors, C, XBee, Linux.

Education

- Oct. 2019 – **Master's Degree, Information Systems**, *Technische Hochschule*, Ulm, Germany
 Mar. 2021 ○ Relevant Coursework: Predictive Analytics, Intelligent and Autonomous Systems, Model-Driven Construction of Complex Systems, Software Quality Assurance, and Information Security.
 ○ Final Grade: 1.9 — Honors: Good
- Sept. 2017 – **Master's Degree, Software Engineering**, *National School of Engineering*, Brest, France
 Mar. 2021 ○ Relevant Coursework: Develop Interactive Applications, UML, Database Management Systems.
 ○ Final Grade: 15.24/20 — Honors: Good
- Mar. 2016 – **Academic Exchange, Computer Engineering**, *Middle East Technical University*, Ankara, Turkey
 Jun. 2017 ○ Relevant Coursework: Programming, Web Development, Database Management Systems, and UML.
 ○ Final Grade: 3.54/4.00 — High Honor
- Sept. 2014 – **Bachelor's Degree, Computer Science and Multimedia**, *Higher Institute of Multimedia Arts*, Manouba, Tunisia)
 Jul. 2017 ○ Relevant Coursework: Mathematics, Programming, Relational Database, and Graphic Design.
 ○ Final Grade: Ranked 1st in promotion — 15.81/20 — Honors: Good

Certificates

- Oct. 2019 – **Certificate in Deep Learning with TensorFlow**, *IBM*
 Nov. 2019 ○ Linear and Logistic Regressions, CNN, RNN, LSTM, and Deployment.
- Jan. 2019 **Certificate in Introduction to Artificial Intelligence**, *Microsoft*
 ○ Machine Learning: Creating a Regression, Classification, and Clustering Models using AzureML Studio.
 ○ Natural Language Processing: Text Analytics, Translation, and Speech.
 ○ Computer Vision: Image Classification, Image Analysis, Face detection, and Recognition.

Skills

Programming Languages	Python, R, Swift, C++, C#, Java, C, Bash, JavaScript
Frameworks	FastAPI, Django, Flask, Celery, Node.js, Express.js, Vue.js, Cutelyst (C++)
Architecture	Microservices, Event-Driven Systems, RESTful APIs, WebSocket, Kafka, OOP
Data Technologies	NoSQL (MongoDB), SQL (PostgreSQL, MySQL), Faust (Stream Processing), JSON
Data Science & ML	TensorFlow, Keras, Scikit-learn, Pandas, Seaborn, AzureML
Cloud & DevOps	Docker, Podman, Kubernetes, Nginx, Linux, CI/CD, Ansible, Git
Quality & Testing	pytest, JMeter, Mypy, Flake8, Black, Unit Testing, Integration Testing, Performance Testing
Methodologies	Agile, Scrum, Trunk-Based Development, Code Review, Mentorship, Pair Programming, Problem Solving
Security & Compliance	ISO 27001, Risk Assessment, Data Security
Tools	Swagger API, Postman API, Jira, Confluence, Slack
Documentation & Design	UML (Modelio), Technical Specifications, System Design Documentation
Languages	Arabic (Native), French (C1), English (TOEIC: 870/990), German (Goethe B1), Turkish (A2)